

2627-MA140 Engineering Calculus

Week 04, Lecture 2 Differentiation Rules

Dr Niall Madden

University of Galway

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Calculus

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$f(x)$	$f'(x)$
x^n	nx^{n-1}
$\ln x$	$\frac{1}{x}$
e^x	e^x
a^{ux}	$ua^{ux} \ln a$
a^x	$a^x \ln a$
$\cos x$	$-\sin x$
$\sin x$	$\cos x$
$\tan x$	$\sec^2 x$
$\cos^{-1} \frac{x}{a}$	$-\frac{1}{\sqrt{a^2 - x^2}}$
$\sin^{-1} \frac{x}{a}$	$\frac{1}{\sqrt{a^2 - x^2}}$
$\tan^{-1} \frac{x}{a}$	$\frac{a}{a^2 + x^2}$

Calculus

Derivatives

Riall an toraidh	$y = uv$ $\Rightarrow \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$	Product rule
Riall an lin	$y = \frac{u}{v}$ $\Rightarrow \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$	Quotient rule
Cuingrial	$f(x) = u(v(x))$ $\Rightarrow f'(x) = \frac{du}{dv} \frac{dv}{dx}$	Chain rule



Let's learn about...

1 Recall: Derivative as a function

2 Differentiation by rule

- 1. The Constant Rule
- 2. The Power Rule
- 3. The constant multiple rule

■ 4. Sum and Difference Rules

3 The Product Rule

4 The Quotient Rule

5 Chain Rule

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7 Exercises

See also Section 3.3 of **Calculus** by Strang & Herman:

<https://openstax.org/books/calculus-volume-1/pages/3-3-differentiation-rules>

Recall: Derivative as a function

Yesterday, we learned that...

Definition: derivative as a function

If $f(x)$ is a be a function, its derivative (function), denoted $f'(x)$, is given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

wherever that limit exists. The process of finding f' is called **differentiation**.

We then used that definition to find the derivative of $f(x) = x^2$, $f(x) = \sqrt{x}$, and $f(x) = \sin(x)$.

Using this “limit” definition to find f' is called “*differentiation from first principles*”. However, we usually use some rules (which all derive from the above definition) in order to find f' .

The Constant Rule

If f is a constant function, i.e. $f(x) = c$ for all x , then $f'(x) = 0$.

Why:

(i) Suppose $f(x)$ represents the velocity of a car. If that velocity is constant (unchanging) so the rate of change is zero.

(ii) If $f(x) = c$ for all x , then

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

We've already deduced that

- ▶ The derivative of $f(x) = x^2$ is $f'(x) = 2x$
- ▶ The derivative of $f(x) = x^{1/2}$ is $f'(x) = \frac{1}{2}x^{-1/2}$

These are particular examples of the **Power Rules**

The Power Rule

If n is any real number, then

$$\frac{d}{dx}(x^n) = nx^{n-1} \Rightarrow \begin{aligned} f(x) &= x^n \\ f'(x) &= nx^{n-1} \end{aligned}$$

Check: (i) If $f(x) = x^{1/2}$, then $n = 1/2$ so

$$f'(x) = \left(\frac{1}{2}\right)x^{(1/2)-1} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}.$$

Examples Calculate the derivatives of the following functions

1. $f(x) = x^6$

2. $f(x) = \sqrt[3]{x}$

1. $f(x) = x^6$ so $n = 6$.
Therefore $f'(x) = nx^{(n-1)} = 6x^5$

2. $f(x) = \sqrt[3]{x} = x^{1/3}$ so $f'(x) = \frac{1}{3}x^{(1/3-1)}$
ie $f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$

The constant multiple rule

Let $f(x)$ be any differentiable function, and let k be constant, then

$$\frac{d}{dx}(kf(x)) = k \frac{d}{dx}(f(x)).$$

Example: Find the derivative of $f(x) = 5x^4$.

$$\begin{aligned}\frac{d}{dx}[5x^4] &= 5 \frac{d}{dx}[x^4] = 5[4x^3] \\ &= 20x^3.\end{aligned}$$

The Sum and Difference Rules

Let $u(x)$ and $v(x)$ be any differentiable functions. Then

$$\frac{d}{dx}(u(x) + v(x)) = \frac{d}{dx}(u(x)) + \frac{d}{dx}(v(x)).$$

$\Rightarrow (u+v)' = u' + v'$

Similarly, $\frac{d}{dx}(u(x) - v(x)) = \frac{d}{dx}(u(x)) - \frac{d}{dx}(v(x)).$

Example: Find the derivative of $f(x) = 1 + x + x^2$.

$$\text{So } f'(x) = (1 + x + x^2)' = (1)' + (x)' + (x^2)'$$

$$= 0 + 1 + 2x$$

Ans:

$$f'(x) = 1 + 2x$$

Actually, the "**Difference Rule**", which states that

$$\frac{d}{dx}(u(x) - v(x)) = \frac{d}{dx}(u(x)) - \frac{d}{dx}(v(x)).$$

can be combined by combining the **Sum Rule** and the **Constant Multiple Rule**.

$$\begin{aligned}(u - v)' &= (u + (-1)v)' \\ &= u' + \underbrace{[(-1)v]'}_{\substack{\text{const} \\ \text{mult} \\ \text{rule}}} = u' + (-1)v' \\ &= u' - v' \quad \checkmark\end{aligned}$$

Example

Suppose that $f(x) = -5x^3 + 3x^2 - 9x + 7$, then find:

- (a) The derivative of $f(x)$;
- (b) The slope of the tangent line at $x = 2$;
- (c) The equation of the tangent at $x = 2$.

(a) $f'(x) = -15x^2 + 6x - 9 + 0$.

- (b) The slope of the tangent line at $x = 2$ is $f'(2)$:

$$f'(2) = -15(2)^2 + 6(2) - 9 = -15(4) + 12 - 9 = -60 + 12 - 9 = -57.$$

- (c) The equation of the line with slope M and passing through a point (x_1, y_1) is

$$y - y_1 = M(x - x_1)$$

$$x_1 = 2, \quad y_1 = f(2), \quad M = f'(2).$$

The y coordinate at $x = 2$ is

$$\begin{aligned} f(2) &= -5(2)^3 + 3(2)^2 - 9(2) + 7 \\ &= -5(8) + 3(4) - 18 + 7 \\ &= -40 + 12 - 18 + 7 \\ &= -39 = y_1 \end{aligned}$$

So the tangent line passes through the point $(2, -39)$ and the slope of the line is -57 .

Therefore, the equation of this line is $y + 39 = -57(x - 2)$

Ans: The equation of the tangent line is $x = 2$ is $y = 75 - 57x$.

The Product Rule

~~$$(uv)' = u'v'$$~~

We now consider some rules which are a little more complicated and, perhaps, less obvious.

The first concerns the derivative of the **product** of two functions.

The Product Rule

Let $u(x)$ and $v(x)$ be any differentiable functions. Then

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}.$$

Let's check that this gives the current answer when evaluating the derivative of $f(x) = x^3$ when we set $u(x) = x^2$ and $v(x) = x$:

1. $f(x) = x^3$ so $f'(x) = 3x^2$ (Power Rule)

2. $f(x) = (x^2)(x) = uv$ where $u = x^2$ $v = x$.
so $u' = 2x$, $v' = 1 \Rightarrow f'(x) = (x^2)(1) + (x)(2x) = x^2 + 2x^2$ ✓

The Product Rule

The Product Rule

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}.$$

Example Use the product rule to find the derivative of $f(x) = x^3(x^2 + 1)$.

1. Without Product Rule:

$$f(x) = x^5 + x^3, \text{ so } f'(x) = 5x^4 + 3x^2.$$

2. With Power Rule,

$$u(x) = x^3$$

$$u'(x) = 3x^2$$

set

$$v(x) = x^2 + 1$$

$$v'(x) = 2x$$

$$\begin{aligned} \text{So } f'(x) &= uv' + v u' = (x^3)(2x) + (x^2 + 1)(3x^2) \\ &= 2x^4 + 3x^4 + 3x^2 \quad \checkmark \end{aligned}$$

The Product Rule

Example: use the product rule to show that, if $f(x) = x \sin(x)$,
then $f'(x) = x \cos(x) + \sin(x)$.

Set $u(x) = x$ $v(x) = \sin(x)$

Then $u'(x) = 1$ $v'(x) = \cos(x)$

So $f'(x) = u v' + v u' = (x) [\cos(x)] + \sin(x)(1)$
 $= x \cos(x) + \sin(x).$

The Quotient Rule

The Quotient Rule

If u and v are differentiable at x and if $v(x) \neq 0$, then $f(x) = \frac{u(x)}{v(x)}$ is differentiable at x , and

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}.$$

Note

$$f(x) = \frac{1}{x} + \frac{1}{x^2} + \dots$$

Example: Use this rule to find the derivative of $f(x) = \frac{x+1}{x^2}$

$$u(x) = x+1 \quad v(x) = x^2$$

$$u'(x) = 1 \quad v'(x) = 2x$$

$$\begin{aligned} \text{so } f'(x) &= \frac{v u' - u v'}{v^2} = \frac{(x^2)(1) - (x+1)(2x)}{(x^2)^2} \\ &= \dots = -x^{-2} - 2x^{-3} \end{aligned}$$

The Quotient Rule

Example

We know that

▶ $\tan(x) = \frac{\sin(x)}{\cos(x)}$, ✓

▶ $\sin^2(x) + \cos^2(x) = 1$ (Fact!) Pythagoras.

▶ $\sin'(x) = \cos(x)$ and $\cos'(x) = -\sin(x)$. (trust me!)

Use these facts, and the Quotient Rule to show that

$$\frac{d}{dx}(\tan(x)) = \left(\frac{1}{\cos(x)}\right)^2.$$

$$f(x) = \tan(x) = \frac{\sin(x)}{\cos(x)}$$

The Quotient Rule

$$\text{so set } u(x) = \sin(x)$$

$$\Rightarrow u'(x) = \cos(x)$$

$$v(x) = \cos(x)$$

$$v'(x) = -\sin(x)$$

Apply Quotient Rule.

$$f'(x) = \frac{v u' - u v'}{v^2} = \frac{\cos(x)\cos(x) - \sin(x)(-\sin(x))}{(\cos(x))^2}$$

$$= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)} \quad \checkmark$$

Chain Rule

Of all the differentiation rules, the **chain rule** is the most important: most other rules are actually just special cases of it. It applies to a “function of a function”

The Chain Rule

If $u(x)$ and $v(x)$ are differentiable, and f is the composite function $f(x) = u(v(x))$, then

$$\frac{df}{dx} = \frac{du}{dv} \frac{dv}{dx}.$$

Chain Rule

The Chain Rule

If $f(x) = u(v(x))$, then

$$\frac{df}{dx} = \frac{du}{dv} \frac{dv}{dx}.$$

Example: What is the derivative of $f(x) = \cos(x^2)$?

Chain Rule

Example

Find $\frac{dy}{dx}$ if $y = (x^3 + 4x^4 + 7)^{12}$.

Example: Let $u(v) = v^{12}$ and $v(x) = x^3 + 4x^4 + 7$, then y is $y = u(v(x))$.

Note that

$$\frac{du}{dv} = 12v^{11} \quad \text{and} \quad \frac{dv}{dx} = 3x^2 + 16x^3.$$

By the Chain Rule we have

$$\frac{dy}{dx} = \frac{du}{dv} \frac{dv}{dx} = 12v^{11}(3x^2 + 16x^3),$$

and therefore

$$\frac{dy}{dx} = 12(x^3 + 4x^4 + 7)^{11}(3x^2 + 16x^3).$$

Chain Rule

Example (Not done in detail in class)

Find $\frac{dy}{dx}$ if $y = \frac{1}{(x^4 + 2x^2 + 8)^{40}}$.

We have $y = (x^4 + 2x^2 + 8)^{-40}$. We can write y as $y(x) = u(v(x))$ with

► $u(v) = v^{-40}$ and so $\frac{du}{dv} = -40v^{-41}$; and

► $v(x) = x^4 + 2x^2 + 8$, so $\frac{dv}{dx} = 4x^3 + 4x$.

Applying the Chain Rule: $\frac{dy}{dx} = \frac{du}{dv} \frac{dv}{dx}$, we get

$$\frac{dy}{dx} = -40v^{-41}(4x^3 + 4x) = \frac{-40(4x^3 + 4x)}{(x^4 + 2x^2 + 8)^{41}}$$

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Calculus

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Derivatives

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$\sin^{-1} \frac{x}{a}$	$\frac{1}{\sqrt{a^2 - x^2}}$
$\tan^{-1} \frac{x}{a}$	$\frac{a}{a^2 + x^2}$

$\frac{1}{\cos^2 x}$

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$$y = uv$$

$$\Rightarrow \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Product rule

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$$y = \frac{u}{v}$$

$$\Rightarrow \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Quotient rule

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$$f(x) = u(v(x))$$

$$\Rightarrow f'(x) = \frac{du}{dv} \frac{dv}{dx}$$

Chain rule

Exercises

Exercises 4.2.1 (Based on Q2(a), 2023/2024)

Find the derivative of $f(x) = \frac{\sin(x)}{\sqrt{x}}$.

Exercise 4.2.2 (Based on Q2(b), 2019/2020)

Find the derivative of $f(x) = \frac{x^2 + x - 2}{x^3 + 6}$.

Exercise 4.2.3

Find the derivative of

1. $f(x) = x^3 \cos(x^2)$
2. $f(x) = \tan^3(\sin^2(x^4))$